

YOUNGSTOWN-WARREN, OH, USA

WMO: 725250

Lat: 41.255N

Lon: 80.674W

Elev: 1167

StdP: 14.09

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	7.1	-5.2	4.3	4.4	-1.0	5.4	8.9	25.4	27.1	23.6	27.8	8.4	240	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	88.1	72.6	85.6	71.2	83.3	69.7	75.0	84.3	73.4	81.6	72.0	79.8	9.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	124.3	79.6	70.7	118.2	77.8	69.4	112.9	76.4	39.2	84.6	37.7	81.7	36.4	79.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.3	18.8	17.0	-3.0	91.5	7.2	2.9	-8.2	93.6	-12.5	95.3	-16.5	97.0	-21.8	99.1	
			DB												
			WB	-3.7	77.8	7.0	1.8	-8.8	79.1	-12.9	80.1	-16.8	81.1	-21.9	82.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	26.3	28.9	37.0	49.1	59.2	67.5	71.2	69.8	63.4	52.3	41.4	32.0
	DBStd	18.20	12.18	11.54	11.50	9.78	8.60	6.77	5.39	5.32	7.13	8.57	9.43	9.87
	HDD50	2847	739	597	433	132	18	0	0	0	2	75	289	562
	HDD65	6113	1200	1012	868	483	223	52	10	16	114	405	708	1022
	CDD50	2829	4	6	30	106	302	525	657	614	403	145	32	5
	CDD65	620	0	0	1	7	43	127	202	165	65	10	0	0
	CDH74	5131	0	0	9	84	413	1071	1713	1271	507	62	1	0
	CDH80	1385	0	0	0	11	84	300	523	337	122	8	0	0
	WSAvg	8.3	10.0	9.6	9.6	9.6	8.1	7.2	6.6	6.4	6.8	8.0	9.0	9.3
	PrecAvg	38.6	2.6	2.2	3.1	3.3	3.7	3.9	4.2	3.4	3.6	2.8	3.0	3.0
Precipitation	PrecMax	54.1	5.9	5.8	6.4	7.3	8.3	10.7	10.4	7.8	7.9	6.1	9.1	6.5
	PrecMin	23.8	0.7	0.6	1.1	1.0	0.8	0.7	0.7	0.5	0.3	0.4	0.8	1.0
	PrecStd	6.0	1.3	1.1	1.2	1.4	1.6	1.9	2.0	1.7	1.7	1.3	1.4	1.1
	PrecStd	6.0	1.3	1.1	1.2	1.4	1.6	1.9	2.0	1.7	1.7	1.3	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	64.8	74.0	81.7	86.1	89.7	91.4	90.3	87.8	80.1	69.5	62.1
		MCWB	55.0	54.6	59.5	62.9	69.3	73.2	74.7	74.2	71.8	67.6	57.9	55.7
	2%	DB	55.9	56.8	66.9	76.1	82.3	86.4	88.2	86.8	83.7	74.7	64.6	56.4
		MCWB	52.5	50.4	54.9	59.7	66.8	71.5	73.1	71.9	69.6	63.6	55.7	51.8
	5%	DB	50.0	50.9	61.2	71.2	78.9	83.6	85.5	83.7	79.8	70.8	60.8	51.7
		MCWB	46.5	45.2	52.0	57.4	65.3	70.0	71.4	70.5	67.9	61.7	53.2	47.8
	10%	DB	43.7	45.2	55.7	66.2	74.9	80.4	82.8	81.0	76.2	66.6	56.7	46.6
		MCWB	40.3	40.1	48.6	54.3	62.7	68.0	69.9	68.9	65.8	59.2	50.5	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	56.9	60.6	65.3	72.0	75.4	77.7	76.8	75.0	70.5	61.5	57.5
		MCDB	59.7	62.6	71.4	76.5	82.7	86.0	87.4	86.5	83.8	77.3	66.7	60.8
	2%	WB	52.9	51.3	57.1	62.0	69.2	73.3	75.4	74.3	72.1	66.7	57.4	52.7
		MCDB	55.8	55.7	64.2	72.5	78.6	83.2	84.8	82.4	80.0	72.0	61.8	55.6
	5%	WB	46.6	45.6	53.3	59.3	67.2	71.7	73.6	72.8	69.9	63.5	54.5	47.9
		MCDB	49.5	50.1	59.8	68.1	76.1	80.9	81.8	79.8	76.7	68.5	59.7	50.6
	10%	WB	40.3	40.6	48.6	56.4	64.9	70.0	72.0	71.2	67.8	60.0	50.9	42.9
		MCDB	43.4	44.8	55.4	64.5	72.8	77.6	79.3	78.0	73.6	65.6	56.1	46.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	15.2	17.1	20.6	20.9	20.0	20.2	19.9	19.9	17.8	14.8	12.0
		MCDBR	18.4	21.8	24.2	26.8	24.6	22.8	22.2	22.3	23.2	22.7	20.2	17.9
		MCWBR	15.9	17.3	15.3	15.4	12.5	11.1	9.8	10.1	11.1	12.7	13.9	15.4
	5% WB	MCDBR	17.5	20.4	21.4	22.9	21.2	19.8	19.1	18.6	19.1	19.1	18.2	17.4
		MCWBR	16.0	17.4	15.2	15.1	11.7	10.3	9.4	9.4	9.7	12.5	14.5	16.4
Clear-Sky Solar Irradiance	taub		0.299	0.329	0.353	0.383	0.421	0.443	0.448	0.435	0.383	0.359	0.316	0.298
	taud		2.437	2.334	2.345	2.305	2.214	2.189	2.194	2.222	2.346	2.411	2.511	2.489
	Ebn at Noon		272	278	284	282	272	265	263	262	270	264	264	263
	Edh at Noon		25	32	36	40	45	46	45	43	35	30	23	22
All-Sky Solar Radiation	RadAvg		444	725	1030	1405	1671	1826	1839	1643	1324	853	556	357
	RadStd		42	86	106	142	148	152	138	87	129	82	80	34

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+99
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+99

Nomenclature: See separate page